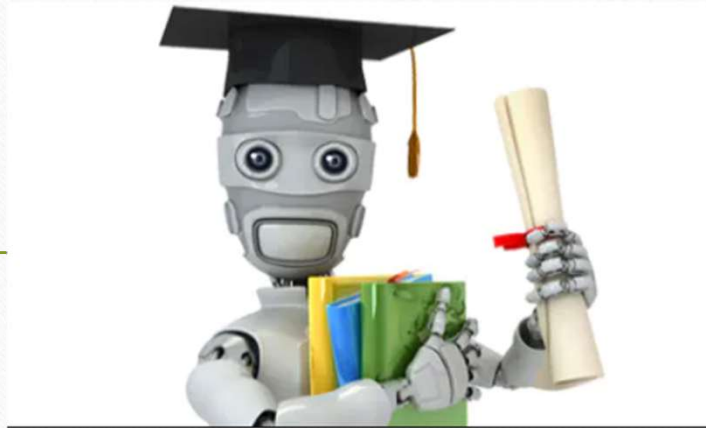


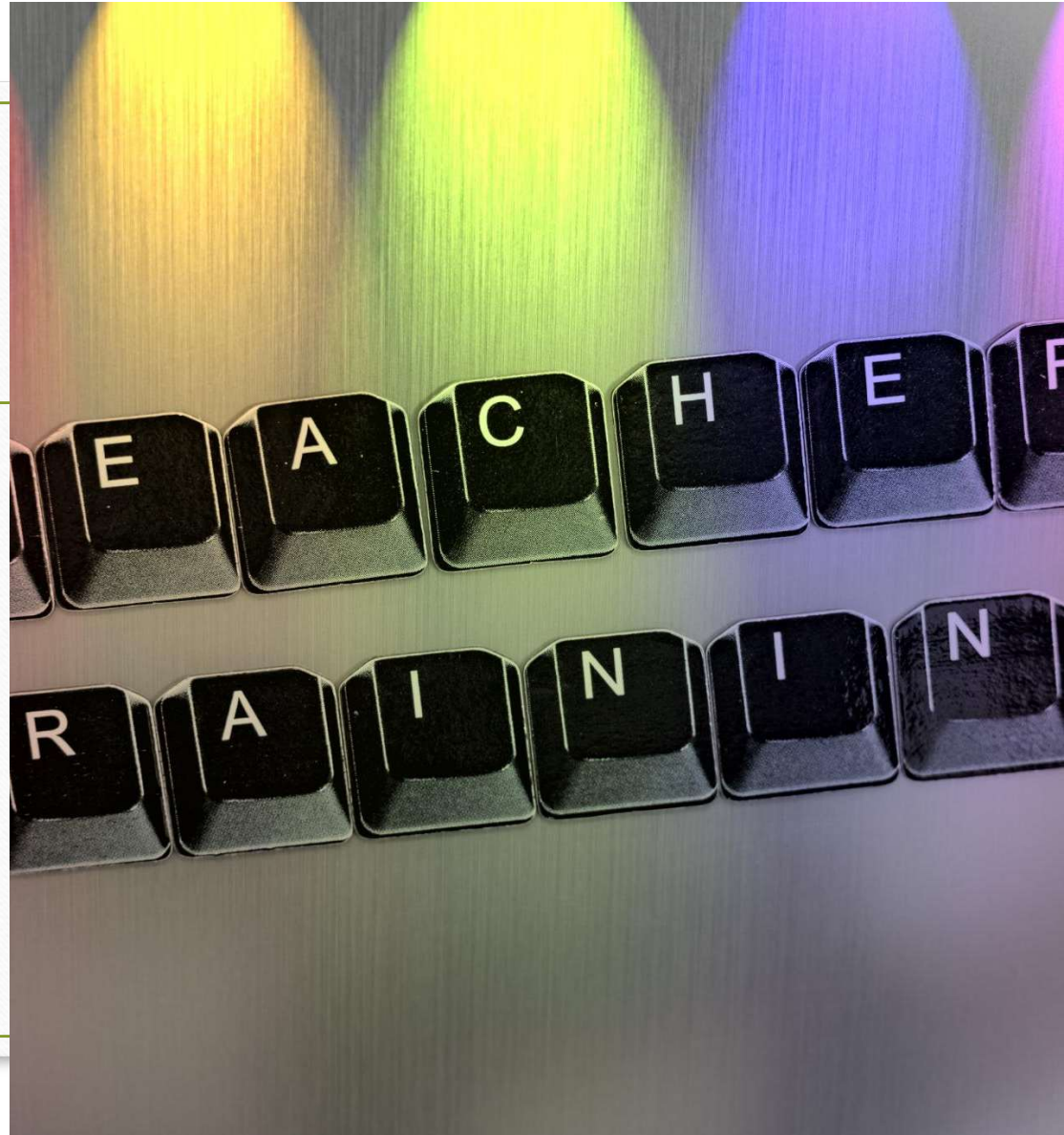


Introduction To Machine Learning

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Agenda

- Introduction to Machine Learning and its Definition
- Exploration of When Machine Learning is Used
- Overview of Types of Learning Techniques
- Steps to Build a Machine Learning Model
- Popular Python Libraries for Machine Learning
- Analysis of the Iris Flower Data Set



What is Machine Learning?

“Learning is any process by which a system improves performance from experience.”

-Herbert Simon

Traditional Programming



Machine Learning



When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise
(speech recognition)
- Models are based on huge amounts of data

Real World Applications of Machine Learning



Face Recognition



Siri and Cortana



Healthcare Industry



Weather Forecasting



Produce a Web Series



Prepare a new Drink

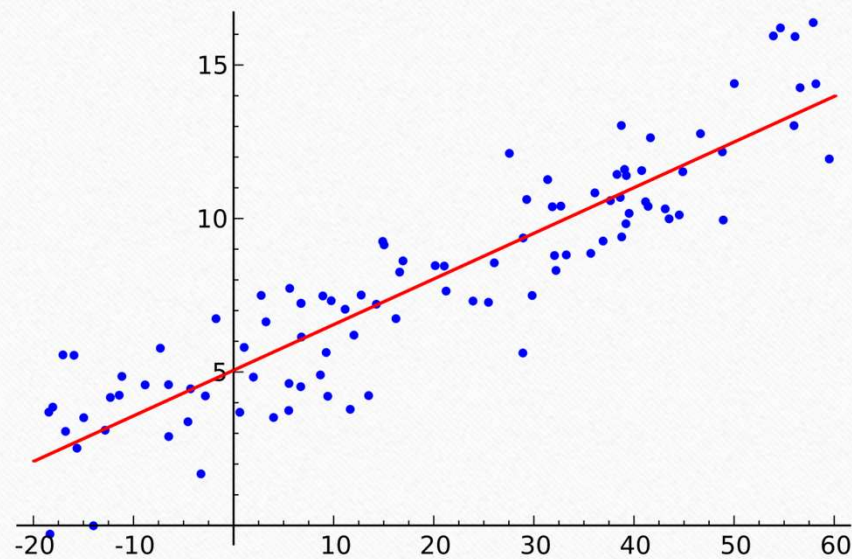
Types of Learning:

- Supervised (inductive) learning—Given: training data + desired outputs (labels).
- Unsupervised learning—Given: training data (without desired outputs).
- Reinforcement learning—Rewards from sequence of actions.

Supervised Learning:

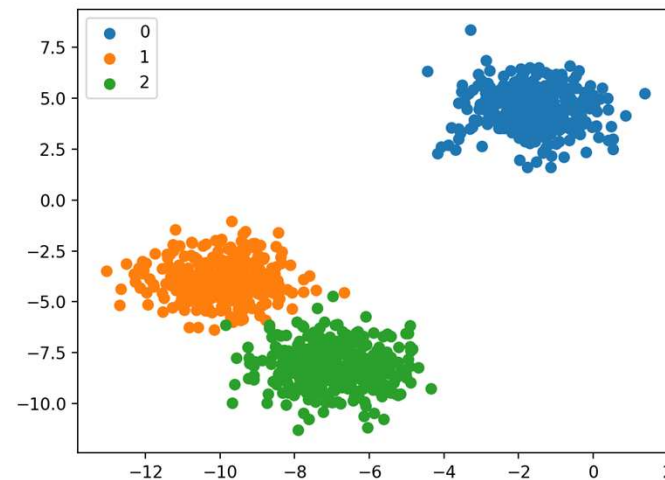
1) Regression

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
 - Learn a function $f(x)$ to predict y given x —y-axis real-valued == regression
-



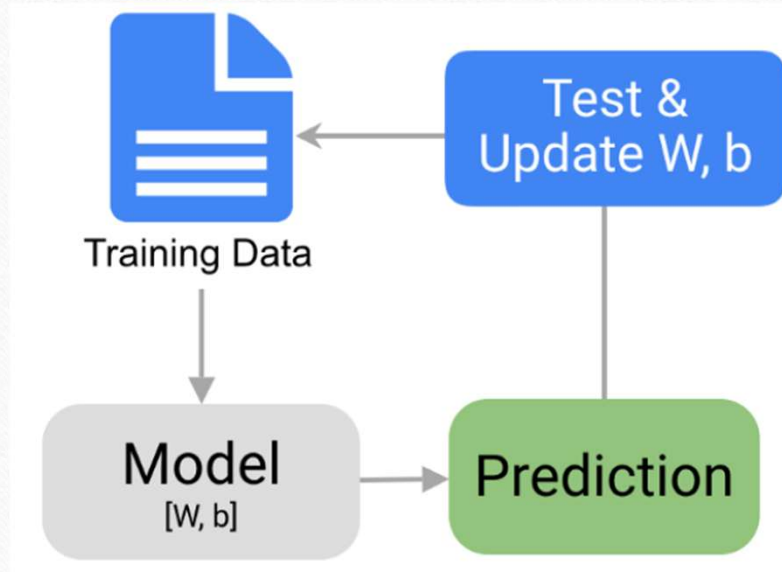
2) Classification

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
 - Learn a function $f(x)$ to predict y given x —y-axis categorical
== classification
-



Steps to build model:

- 1 - Data Collection
- 2 - Data Preparation
- 3 - Choose a Model
- 4 - Train the Model
- 5 - Evaluate the Model
- 6 - Make Predictions



Python libraries used in Machine Learning:

- 1)Numpy
- 2)Pandas
- 3)Scikit-learn
- 4)Matplotlib
- 5)TensorFlow
- 6)Keras
- 7)PyTorch
- 8)Scipy
- 9)Seaborn

Iris Flower Data Set

Made By:

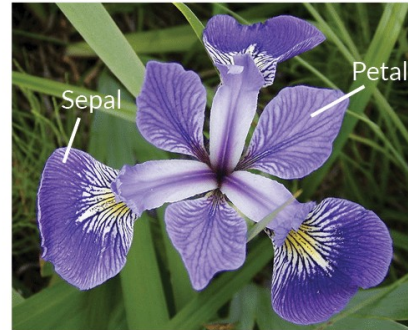
Name: Shivji

Course : B.Tech



- The *Iris* flower data set or Fisher's *Iris* data set is a multivariate data set introduced by Ronald Fisher in his 1936.

It is sometimes called Anderson's *Iris* data set because Edgar Anderson collected the data to quantify the morphologic variation of *Iris* flowers of three related species. The use of this data set in cluster analysis is however uncommon, since the data set only contains two clusters with rather obvious separation.



Iris Versicolor



Iris Setosa



Iris Virginica

- One of the clusters contains *Iris setosa*, while the other cluster contains both *Iris virginica* and *Iris versicolor* and is not separable without the species information Fisher used.

It is multivariate (more than 2 dependent variable) data set Study of three related Iris flowers species. Data set contain 50 sample of each species (Iris-Setosa, Iris-Virginica, IrisVersicolor)

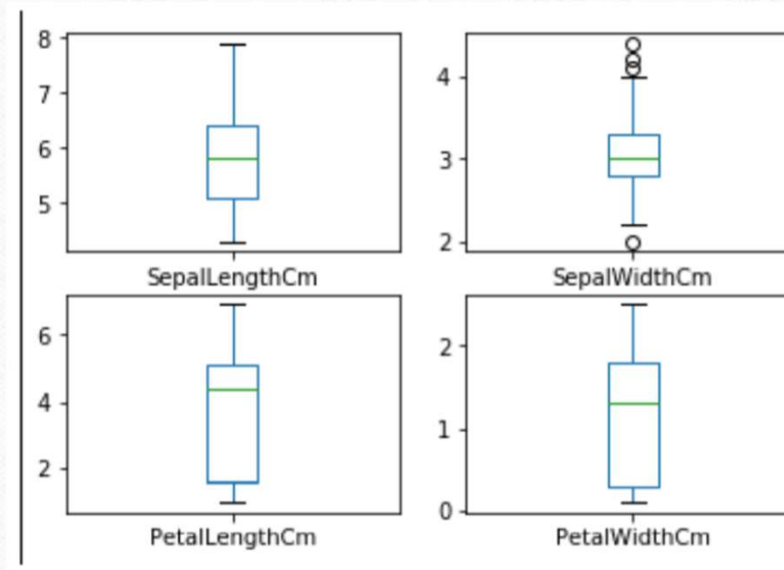
Features Used :

1. Sepal length in cm
2. Sepal width in cm
3. Petal length in cm
4. Petal width in cm

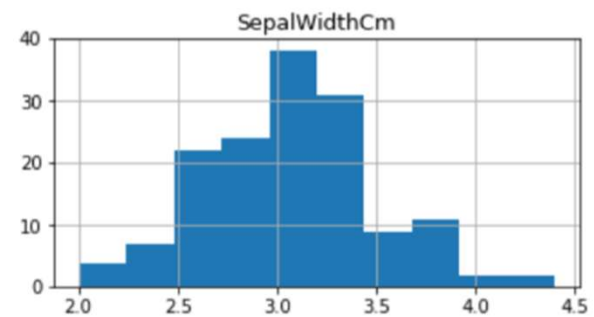
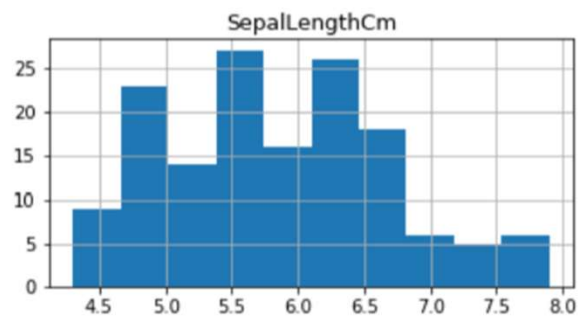
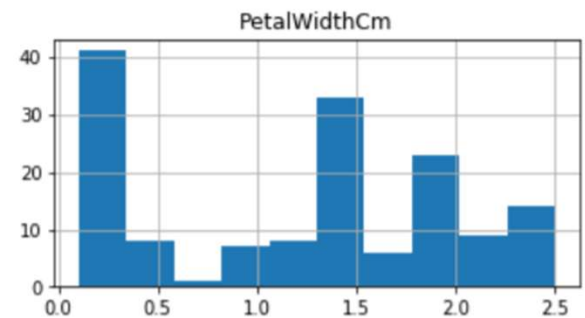
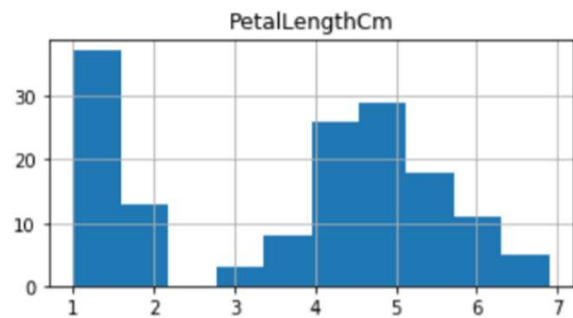
Data Analysis :

1. Descriptive statistics- SD, Min, Max etc .
2. Class Distribution (Species counts are balanced or imbalanced) – Balanced.
3. Univariate Plots:- Understand each attribute better.

Box and whisker plots(Give idea about distribution of input attributes)

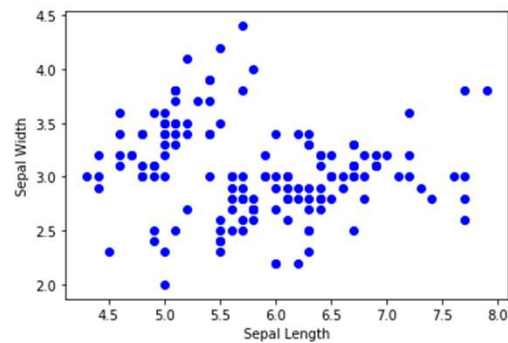


Plotting Histogram:



Plotting Scatter Graph Between Sepal Length and Sepal Width:

```
In [75]: plt.xlabel("Sepal Length")  
plt.ylabel("Sepal Width")  
plt.scatter(X,Y,color='b')  
plt.show()
```



Observation:

-
1. By utilizing the Sepal_Length and Sepal_Width features, we can effectively differentiate the Setosa flower from the other types.
 2. Distinguishing between Versicolor and Virginica is significantly more challenging due to their considerable overlap.
 3. Therefore, the Sepal_Length and Sepal_Width features are primarily effective for identifying Setosa.

Implementation of Machine Learning.



Steps to implement Machine Learning

1. Import Library
2. Analyze Data
3. Splitting the Data Set into train and test
4. Chossing right algorithm for training model
5. Test the algorithm with test data.

Algorithms Implemented:

1. Logistic Regression
2. Support Vector Machine
3. Classification and Regression Tree (CART)
4. Gaussian Naive Bayes (NB)
5. K-Nearest Neighbors (KNN)
6. Decision Tree

Final Evaluation Of All Models:

```
In [40]: results = pd.DataFrame({
        'Model': ['Logistic Regression', 'Support Vector Machines', 'Naive Bayes', 'KNN', 'Decision Tree'],
        'Score': [0.947, 0.947, 0.947, 0.947, 0.921]})

result_df = results.sort_values(by='Score', ascending=False)
result_df = result_df.set_index('Score')
result_df.head(9)
```

Out[40]:

	Model
Score	
0.947	Logistic Regression
0.947	Support Vector Machines
0.947	Naive Bayes
0.947	KNN
0.921	Decision Tree

THANK
YOU